

Week 8 Progress Report

Project: Room 315 kinematic shuttle simulation
Reporting period: April 21, 2026 to April 28, 2026

This week focused on completing the user-facing control and monitoring layer for the Room 315 shuttle cell. The work concentrated on sensor publication, naming alignment, runtime shuttle recovery, and report-ready engineering figures generated from the calibrated rail geometry.

1. Main Progress Achieved

- User-facing switch, segment, stopper, and detector labels were aligned with the current Room 315 report layout.
- A complete right-rail virtual sensing layer was integrated through dedicated ROS 2 topics for approach events and position events.
- Runtime shuttle management was extended so a shuttle can be reset, removed, and re-added without restarting Gazebo.
- Report-ready figures were generated from the Room 315 CSV geometry and cubic Hermite path, including sensor, slot, and stopper placement.
- The README files were rewritten so they describe only the current interface and current published names.

2. Technical Outcome of the Week

The Room 315 shuttle node now publishes a consistent operator interface: the route command topic, the state topic, the stopper workflow topic, and the position detector topic all expose the same naming scheme used in the final diagrams. This reduces confusion during Gazebo testing, topic monitoring, and documentation review.

A second outcome is better runtime robustness. The operator can recover from a bad route selection by sending `RESET`, can remove a shuttle model completely by sending `REMOVE`, and can add a new shuttle again through the runtime add topic. This removes the need to stop and relaunch Gazebo during iterative tests.

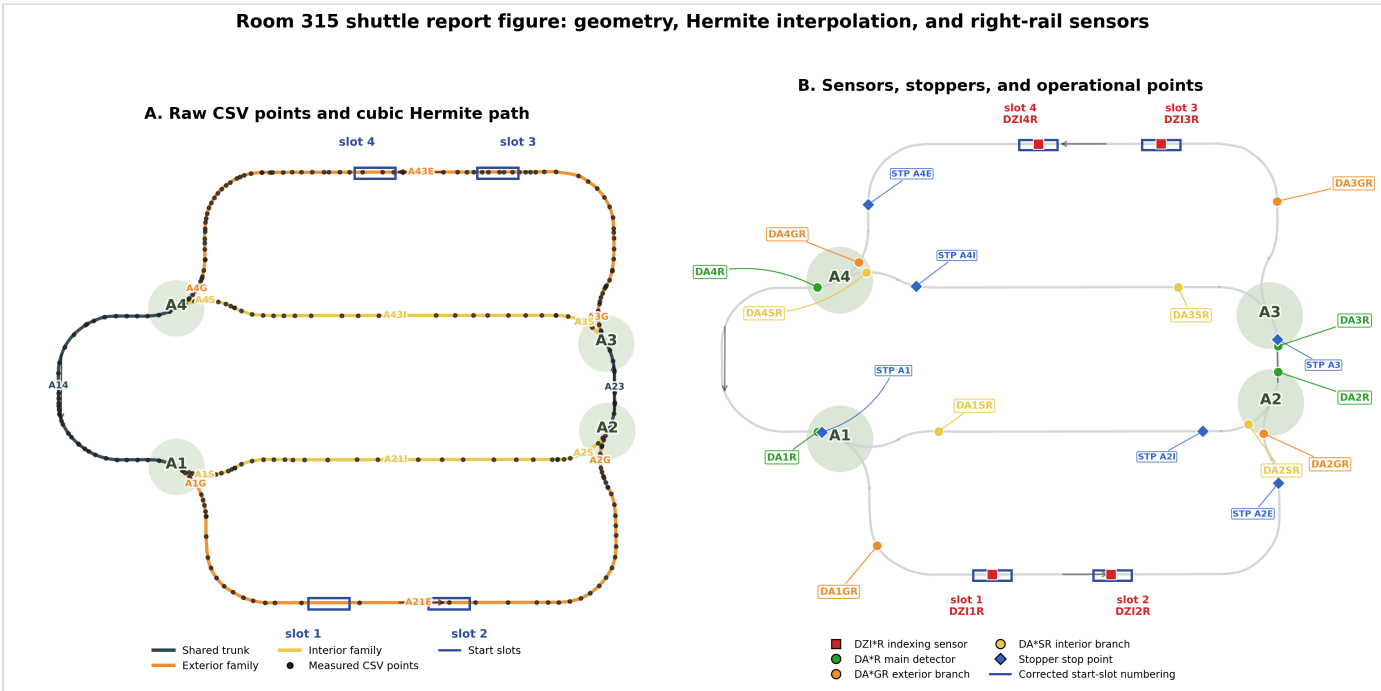


Figure 1. Week 8 summary figure showing the Room 315 geometry, cubic Hermite path interpolation, start slots, right-rail sensors, and stopper locations used by the current software interface.

3. Sensor System Added This Week

Two sensing layers are now available. The first layer is the `/room_315/sensors/switch_approach` topic, which reports when a shuttle is approaching a stopper before a switch. The second layer is the `/room_315/sensors/position` topic, which reports virtual position detectors placed around indexing zones and switch branches on the right rail.

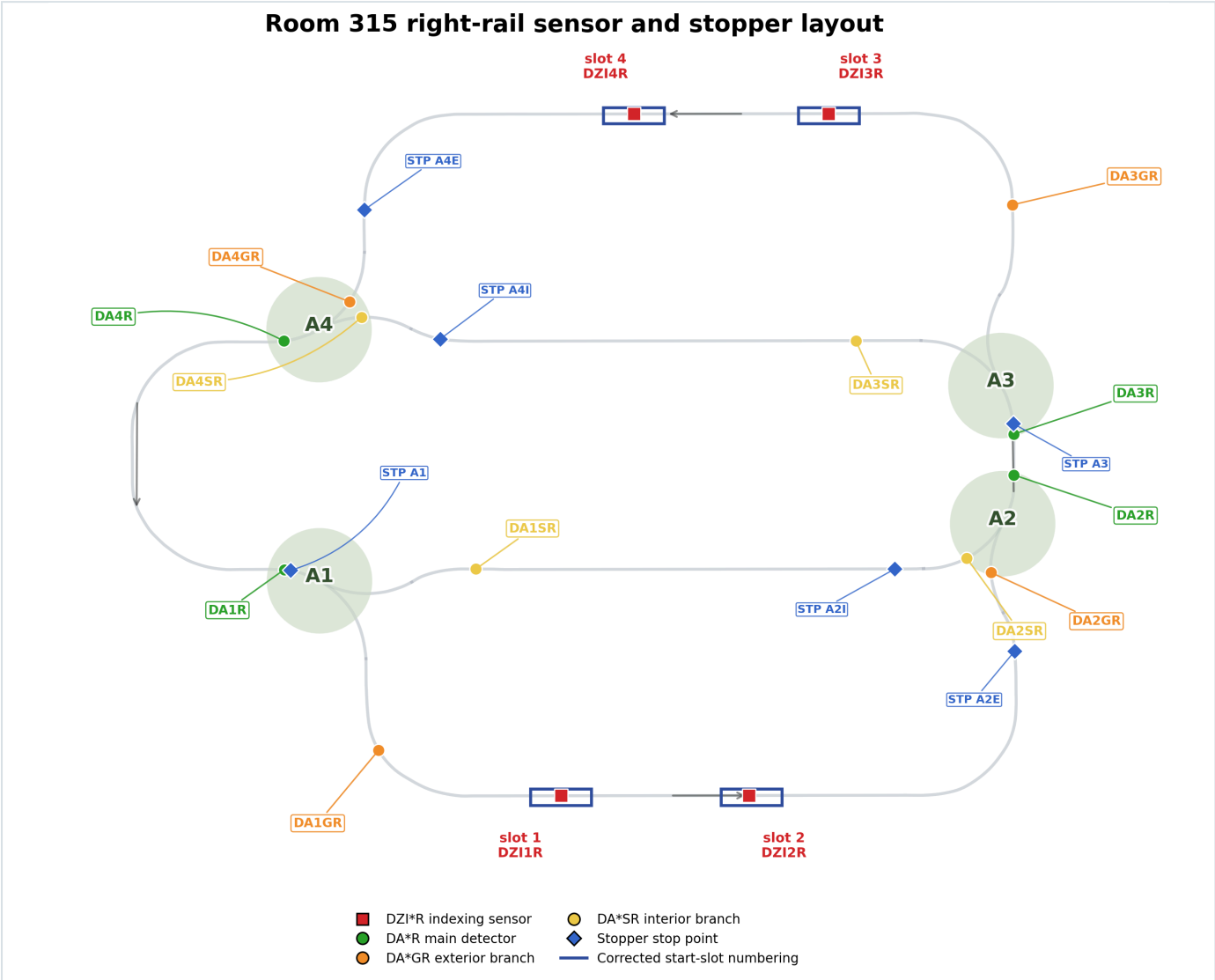


Figure 2. Final right-rail sensor and stopper layout used for reporting and for the published ROS 2 naming convention.

3.1 Indexing-Zone Detectors

Sensor	Location	Function
DZI2R	Slot 1 indexing position	Confirms that the shuttle is located at the first indexing/start position on the right rail.
DZI1R	Slot 2 indexing position	Confirms that the shuttle is located at the second indexing/start position.
DZI4R	Slot 3 indexing position	Confirms shuttle presence at the third indexing/start position.
DZI3R	Slot 4 indexing position	Confirms shuttle presence at the fourth indexing/start position.

3.2 Switch-Area Detectors

Sensor group	Function
DA1R , DA1GR , DA1SR	DA1R detects the shuttle on the common track near switch A1. DA1GR detects the shuttle on the exterior branch of A1. DA1SR detects the shuttle on the interior branch of A1.
DA2R , DA2GR , DA2SR	DA2R detects the shuttle on the common track near switch A2. DA2GR detects the shuttle on the exterior branch of A2. DA2SR detects the shuttle on the interior branch of A2.
DA3R , DA3GR , DA3SR	DA3R detects the shuttle on the common track near switch A3. DA3GR detects the shuttle on the exterior branch of A3. DA3SR detects the shuttle on the interior branch of A3.
DA4R , DA4GR , DA4SR	DA4R detects the shuttle on the common track near switch A4. DA4GR detects the shuttle on the exterior branch of A4. DA4SR detects the shuttle on the interior branch of A4.

4. Commands Added and Their Function

Topic / command	Example	Function
/room_315/switch_states	A1=EXTERIOR , A1=INTERIOR , A1=E , A1=I	Sets the commanded route at a switch using the current operator naming. The same topic also updates the Gazebo switch visuals.
/room_315/switch_states	ALL=EXTERIOR , ALL=INTERIOR	Applies one route selection to all switches at once for fast route testing.
/room_315/stopper_states	A1=1 , A1=0 , ALL=1 , ALL=0	Closes or opens a stopper independently from the switch state to perform safe manual switch handling.
/room_315/shuttle/control_cmd	room315_shuttle_1=OFF , ON	Stops or resumes the kinematic motion of one shuttle while keeping the model available in the simulation.
/room_315/shuttle/control_cmd	room315_shuttle_1=RESET	Returns a shuttle to its configured start slot and recovers it from invalid routing states without restarting Gazebo.
/room_315/shuttle/control_cmd	room315_shuttle_1=REMOVE	Deletes the shuttle model from Gazebo and unregisters it from the running node.
/room_315/shuttle/add_cmd	entity=room315_shuttle_1 slot=1 speed=0.2	Adds a shuttle dynamically at runtime on a selected slot, with an optional custom speed and entity name.
/room_315/sensors/position	ros2 topic echo /room_315/sensors/position	Displays the virtual indexing and switch-area detector events for monitoring and validation.
/room_315/sensors/switch_approach	ros2 topic echo /room_315/sensors/switch_approach	Displays approach events before switches, including the associated stopper and the remaining distance to the stop point.
/room_315/shuttle/state	ros2 topic echo /room_315/shuttle/state grep current_segment	Shows the current user-facing segment label, together with shuttle state, slot, and motion information.

5. Validation and Outcome

- The report figures were regenerated from the current Room 315 geometry and sensor layout.
- The shuttle node was updated so the published names for switches, segments, and detectors match the final report convention.
- The documentation was cleaned to describe only the current interface used in Gazebo and ROS 2 testing.
- The cell can now be tested through a complete operator workflow: route command, stopper control, sensor monitoring, shuttle reset, shuttle removal, and runtime shuttle addition.